Docker Registry

1. What is a Docker Registry?

- A **Docker Registry** is a storage and distribution system for **Docker Images.**
- You can think of it like **GIT** for **Docker Images**
  - Git -> Stores and versions source code
  - Docker Registry -> stores and versions **Docker Images**
- **Docker Images** can have different **tags/versions**

- Example:-

```
web-server
├── 1.0
├── 2.0
└── 3.0
```

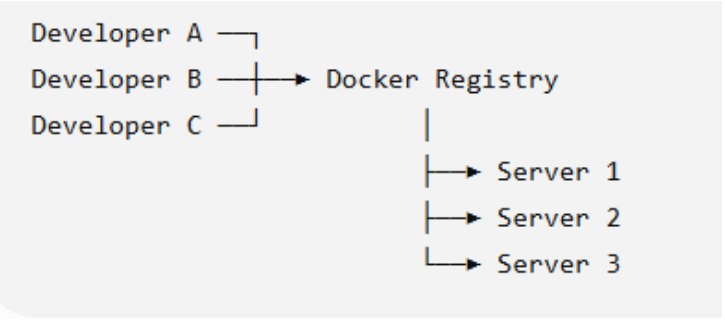
- Images can be:
  - Pulled from the registry
  - Deployed onto servers
- When a new Image version is available, developers/team members can pull and deploy it.

2. Why do we need a Docker Registry?

1. Centralized Storage
2. Versioning
3. Sharing Images

A) Centralized Storage

- A registry acts as a **centralized resource** for **Docker Image.**
- Useful when:
  - Multiple developers are working on a team.
  - Multiple servers need the same images.
  - Servers are located in different locations.
- Instead of manually sharing the image files, everyone can access the image from the registry.



B) Versioning

- Docker registries make it easy to maintain **multiple versions** of the **same Image.**
- Each version can have a different tag.
- This makes it possible to:
  - Deploy a specific version
  - Quickly **roll back to previous version** if required.

- Example

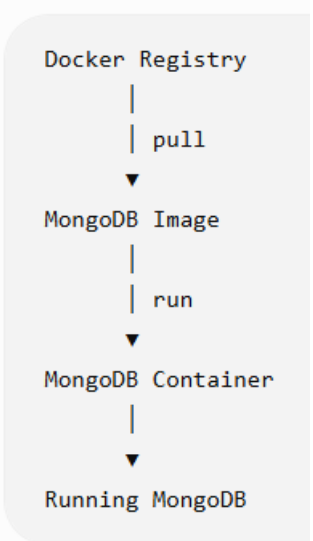
```
my-app:1.0
my-app:2.0
my-app:3.0
```

C) Sharing Images

- Registries can be used to share images publicly.
- This is useful for **open-source** projects.
- Public images can be downloaded and used by other Docker uses.

3. Example: Database Images

- Popular database such as
  - PostgreSQL
  - MongoDB
- Have official Docker images available.
- If Docker is installed on your machine we can:
  1. Pull the database image
  2. Run the image
  3. Get a running database container
- **This means we can run a database without installing the database software directly on our machine.**



4. Docker Hub

- **Docker Hub** is the default registry commonly used by Docker CLI and Docker Tools.
- It hosts a very large **collection** of **public Docker Images.**
- **Images** are provided by
  - Individual developers
  - Software vendors
  - Large corporations
- Docker Hub is one of the most important registries for Docker Developers

- Example of Images available include:
  - Nginx
  - MongoDB
  - Alpine
  - Node.js
  - Redis
  - Python
  - We can search for images directly on Docker Hub
  - We can also create our own Docker Hub Account
  - Images can have multiple **tags/versions**
  - **Docker Hub** provides commands showing how to pull particular versions
  - **Official Images** are marked as Official Images.

5. Other Docker Registries

Amazon ECR

- **Amazon ECR = Amazon Elastic Container Registry**
- Provided by **AWS**
- Useful when your infrastructure is already running on AWS
- Integrates with other AWS services
- Allows you to store Docker/container images within the AWS ecosystem

Google Artifact Registry

- **Google Container Registry / Artifact Registry**
- Provided by **google cloud**
- Allows you to store **Docker Images**
- Integrates with Google Cloud Services
- Useful when you are using google cloud infrastructure

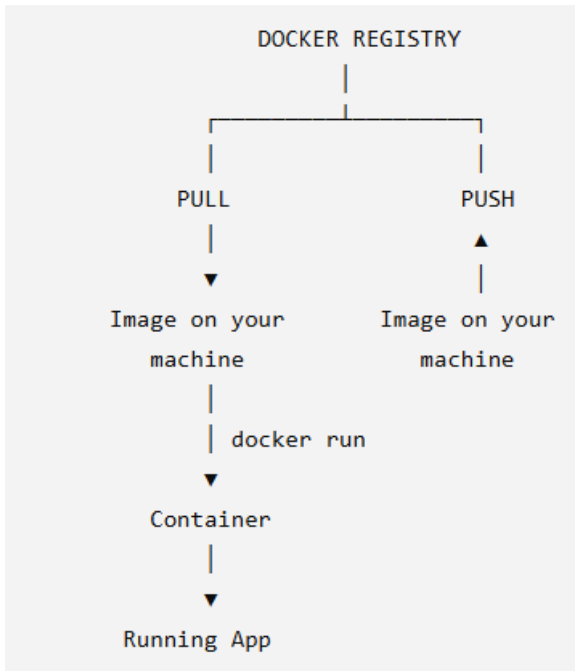
Microsoft Azure Container Registry

- **Azure Container Registry (ACR)** is Microsoft's container registry.
- Useful when you're using **Microsoft Azure.**
- Allows you to stay within the Azure ecosystem.

Quay

- **Quay** is another container registry provided by **Red Hat.**
- It is another option for storing and distributing container images

Registry	Provider/Ecosystem	Typical Use
Docker Hub	Docker	General/public Docker images
Amazon ECR	AWS	AWS infrastructure
Artifact Registry	Google Cloud	Google Cloud infrastructure
Azure Container Registry	Microsoft Azure	Azure infrastructure
Quay	Red Hat	Container image storage/distribution


DOCKER REGISTRY
PULL
PUSH
Image on your machine
Image on your machine
docker run
Container
Running App

Interview – Ready Answer

- A Docker Registry is a centralized storage and distribution system for Docker images.
- It allows teams to store, version, share, pull, and push Docker images.
- Docker Hub is the most commonly used public registry,
- while cloud providers such as AWS, Google Cloud, and Azure provide their own registries for their respective ecosystems.